# MILITARY STANDARD

**MIL-STD-1320-172**
**(NAVY)**

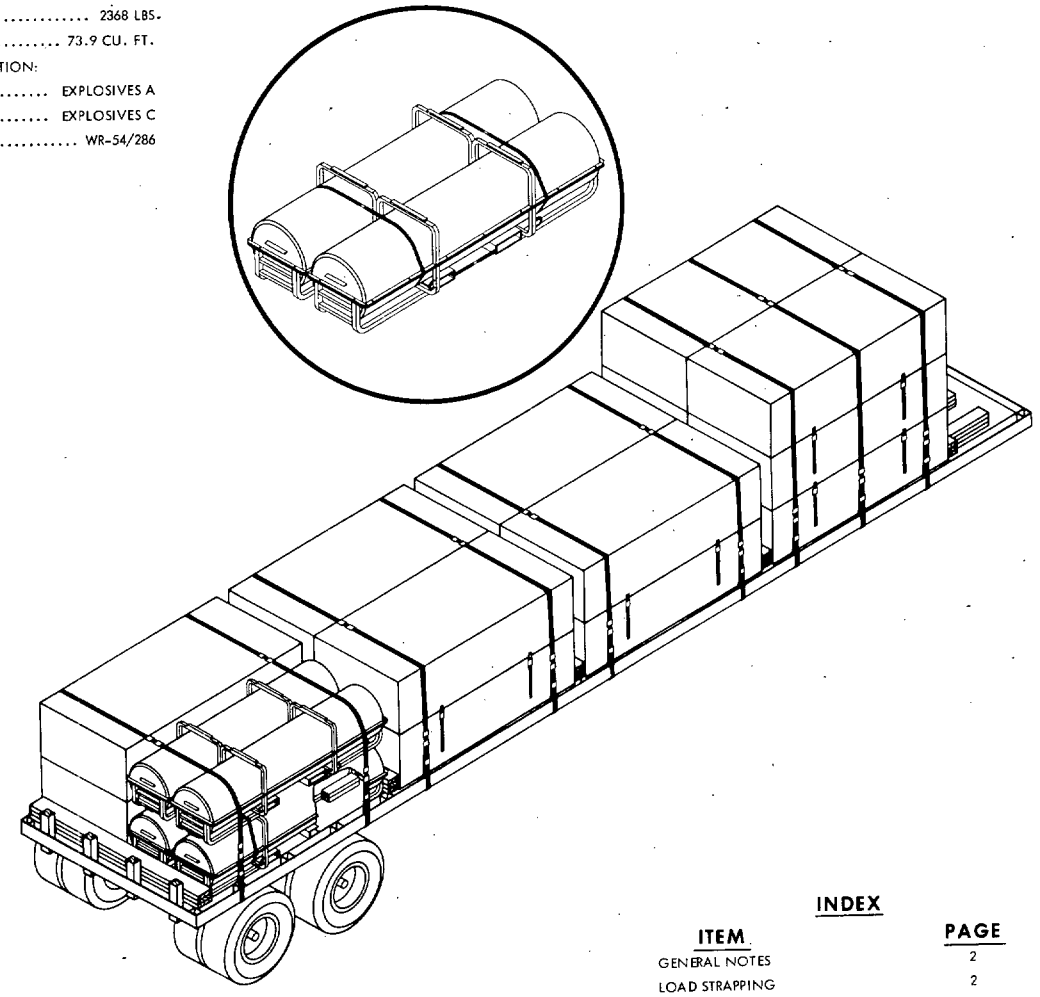
12 JANUARY 1976

## TRUCKLOADING

**DISPENSER AND BOMB, AIRCRAFT, CBU-59/B, COMPLETE (APAM)**
**DISPENSER AND BOMB, AIRCRAFT, CBU-59(T-1)/B, TRAINING**
**IN CONTAINER MK 427 MOD 1**
**FLEET ISSUE UNIT LOAD**

### UNIT LOAD DATA

DIMENSIONS ..... L-103, W-47, H-52
WEIGHT (APPROX) ..... 2368 LBS.
CUBE ..... 73.9 CU. FT.
DOT HAZARD CLASSIFICATION:
CBU-59/B ..... EXPLOSIVES A
CBU-59/(T-1)/B ..... EXPLOSIVES C
UNIT LOAD DWG ..... WR-54/286



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS. REFER TO INDEX TO STANDARDS, MIL-HDBK-236. (NAVY).

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FSC 8140

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SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

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ORIGINATOR Charles M Black 12/15/75
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY



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# GENERAL NOTES

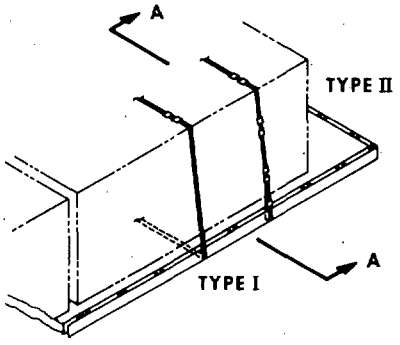
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE DISPENSER AND BOMB, AIRCRAFT CBU-59/B, COMPLETE (APAM) AND CBU-59(T-1)/B, TRAINING, IN CONTAINER MK 427 MOD 1. THE CONTAINERS SHALL BE UNITIZED IN ACCORDANCE WITH WR-54/286.
2. PROCEDURE AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR A 40 FT FLATBED TRAILER. THE TRAILER'S TANDEM AXLES SHALL BE LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. WHEN THE UNIT LOADS ARE STACKED TWO (2) OR MORE HIGH, THEY SHALL BE STACKED AND STRAPPED AS SHOWN ON PAGE 6.
4. SLEEPER, PIECE NO. 3, MUST BE PREPOSITIONED. IT SHOULD BE LOCATED AGAINST THE INSIDE EDGE OF THE INBOARD RUNNER OF THE CONTAINER AS SHOWN IN SECTION C-C.
5. HOLD-DOWN STRAPS SHALL BE 2 X .050 AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENT SHOWN BELOW.

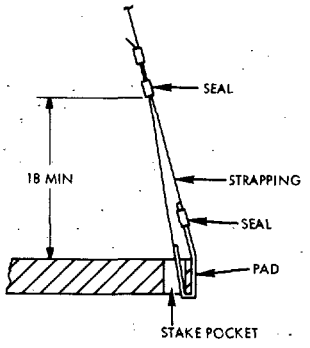
**NOTE**

CHAINS AND LOAD BINDERS ARE NOT AUTHORIZED FOR THIS LOAD AS THEY WILL CAUSE THE LOAD TO EXCEED 96 INCHES IN WIDTH.

6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH APPROPRIATE PLACARD (SEE UNIT LOAD DATA, PAGE 1) TO BOTH SIDES, FRONT, AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS FF-N-105, TYPE II, SYTLE 10, COMMON BRIGHT; STRAPPING QQ-S-781, TYPE I, CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.

# LOAD STRAPPING




**SECTION A-A**

TIE-DOWN STRAPPING

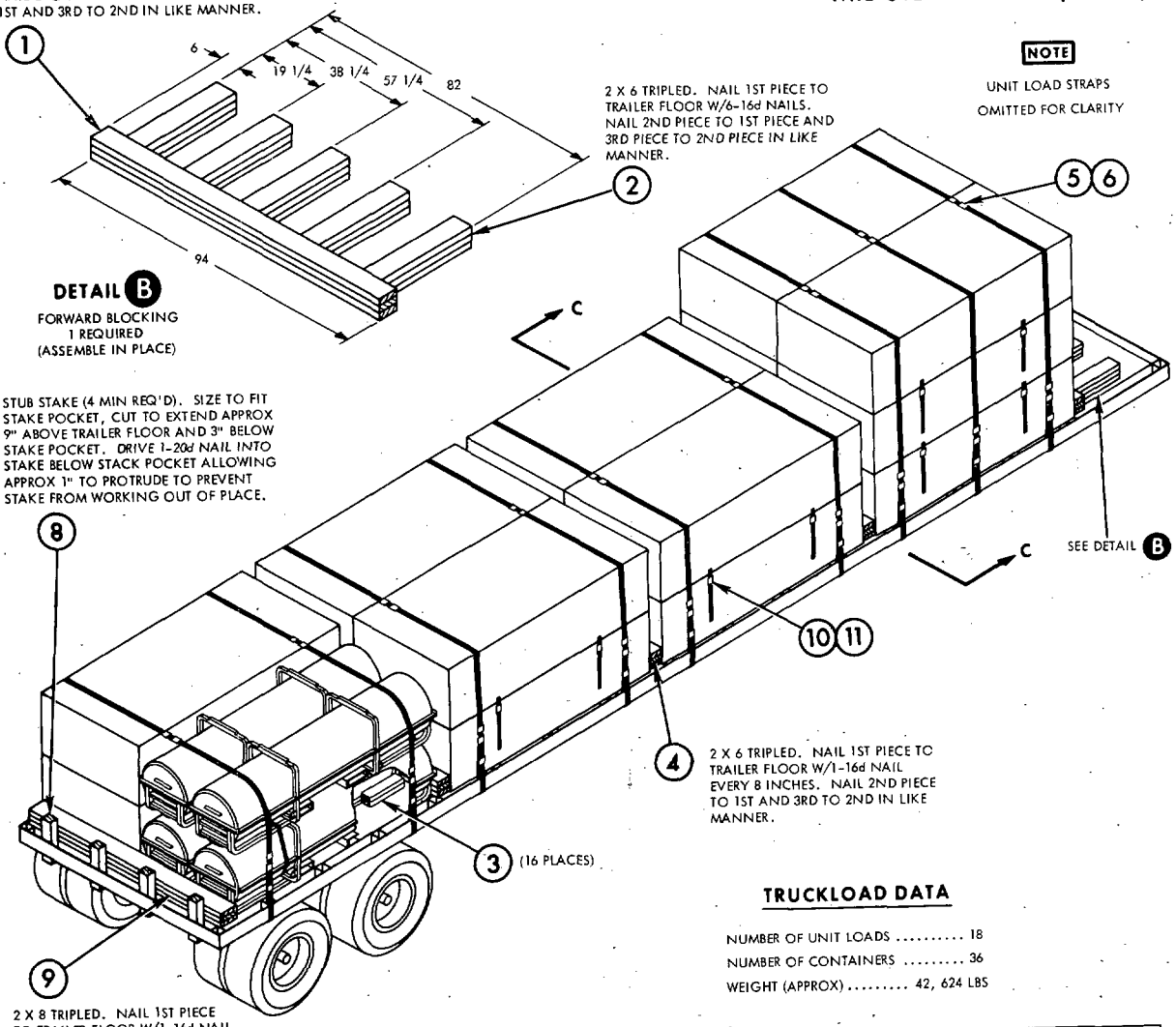
THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAPPING SEALS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PERCENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION, AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD, IF PRACTICABLE.

TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD.

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH TWO DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION A-A. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD (A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

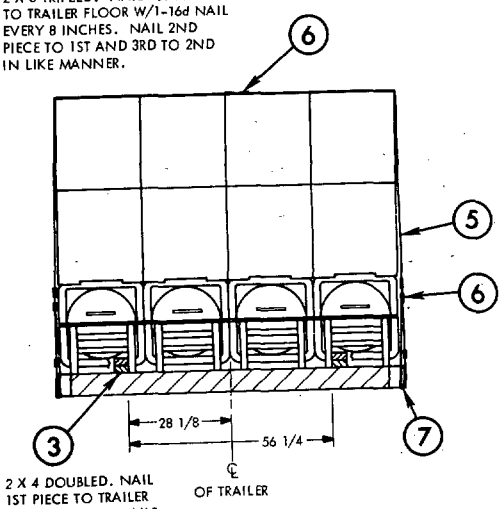


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## TRUCKLOAD DATA

NUMBER OF UNIT LOADS ..... 18
NUMBER OF CONTAINERS ..... 36
WEIGHT (APPROX) ..... 42, 624 LBS




SECTION C-C

| PIECE NO. | DESCRIPTION | SIZE | NO. OF PIECES REQ'D | NAIL TO | NUMBER | SIZE |
|---|---|---|---|---|---|---|
| 11 | SEAL | FOR 1 1/4 IN. STRAP | 20 | - | - | - |
| 10 | UNITIZING STRAP | 1 1/4 X .035 X 13 FT 0 | 20 | SEE PAGE 7 | | |
| 9 | AFT CROSS MEMBER | 2 X 8 X 92 | 3 | SEE FIELD NOTE | | |
| 8 | STUB STAKE | TO SUIT | 4 MIN. | SEE FIELD NOTE | | |
| 7 | STAKE POCKET PAD (STRAPPING) | 2 X .050 X 24 | 18 | - | - | - |
| 6 | SEAL | FOR 2 INCH STRAP | 72 | - | - | - |
| 5 | TIE DOWN STRAP | 2 X .050 X LENGTH TO SUIT | 9 | SEE LOAD STRAPPING, PAGE 2 | | |
| 4 | SEPARATOR | 2 X 6 X 92 | 9 | SEE FIELD NOTE | | |
| 3 | SLEEPER | 2 X 4 X 18 | 32 | SEE SECTION B-B | | |
| 2 | CLEAT | 2 X 6 X 30 | 15 | SEE DETAIL A | | |
| 1 | FWD CROSS MEMBER | 2 X 6 X 92 | 3 | SEE DETAIL A | | |

LIST OF MATERIALS & NAILING DATA



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# LTL 40 FT TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I AND POSITION THE UNIT LOADS ON THE TRAILER ACCORDINGLY.

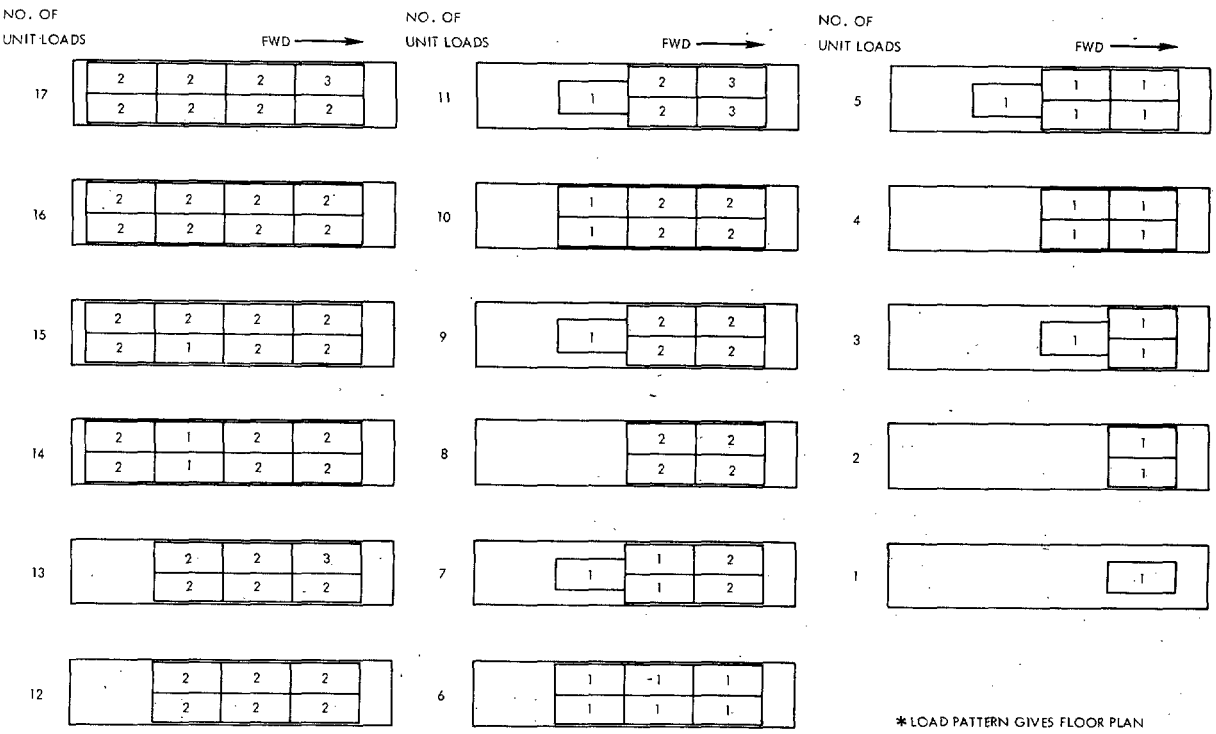
**NOTE**

STACKS OF CONTAINERS 3 HIGH REQUIRE 3 HOLD-DOWN STRAPS. ALL OTHERS REQUIRE 2 HOLD-DOWN STRAPS.

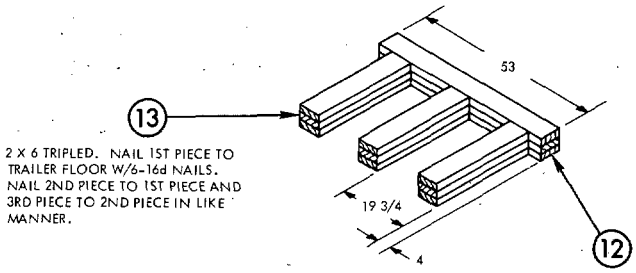
2. WHEN LOADING 14 TO 17 UNIT LOADS, BLOCK, BRACE AND TIE DOWN THE LOAD USING THE PRINCIPLES SHOWN ON PAGE 3.
3. WHEN LOADING 13 OR LESS UNIT LOADS, BLOCK, BRACE AND TIE DOWN THE LOAD USING THE PRINCIPLES SHOWN ON PAGE 5. LTL LOADS OF 2, 4, 6, 8, 10, 12 AND 13 UNIT LOADS SHALL BE BLOCKED AT THE REAR OF THE LOAD USING DETAIL B, PAGE 2.
4. THE LTL ON PAGE 5 IS FOR 11 UNIT LOADS AND SHOWS THE CORRECT BLOCKING, BRACING AND TIE DOWNS FOR A LOAD OF THIS QUANTITY.
5. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED. THE LOAD PATTERN, BLOCKING, BRACING AND TIE DOWNS SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS MILITARY STANDARD AND THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

## TABLE I

LOAD PATTERN *




* LOAD PATTERN GIVES FLOOR PLAN FOR TRAILER. NUMBER IN BLOCKS GIVES NUMBER OF LAYERS OF UNIT LOADS.




(13) 2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

(12) 2 X 6 TRIPLED. NAIL TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.

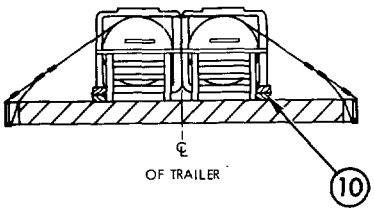
**DETAIL D**

REAR BLOCKING ASSEMBLY

(SEE LTL NOTE 3)



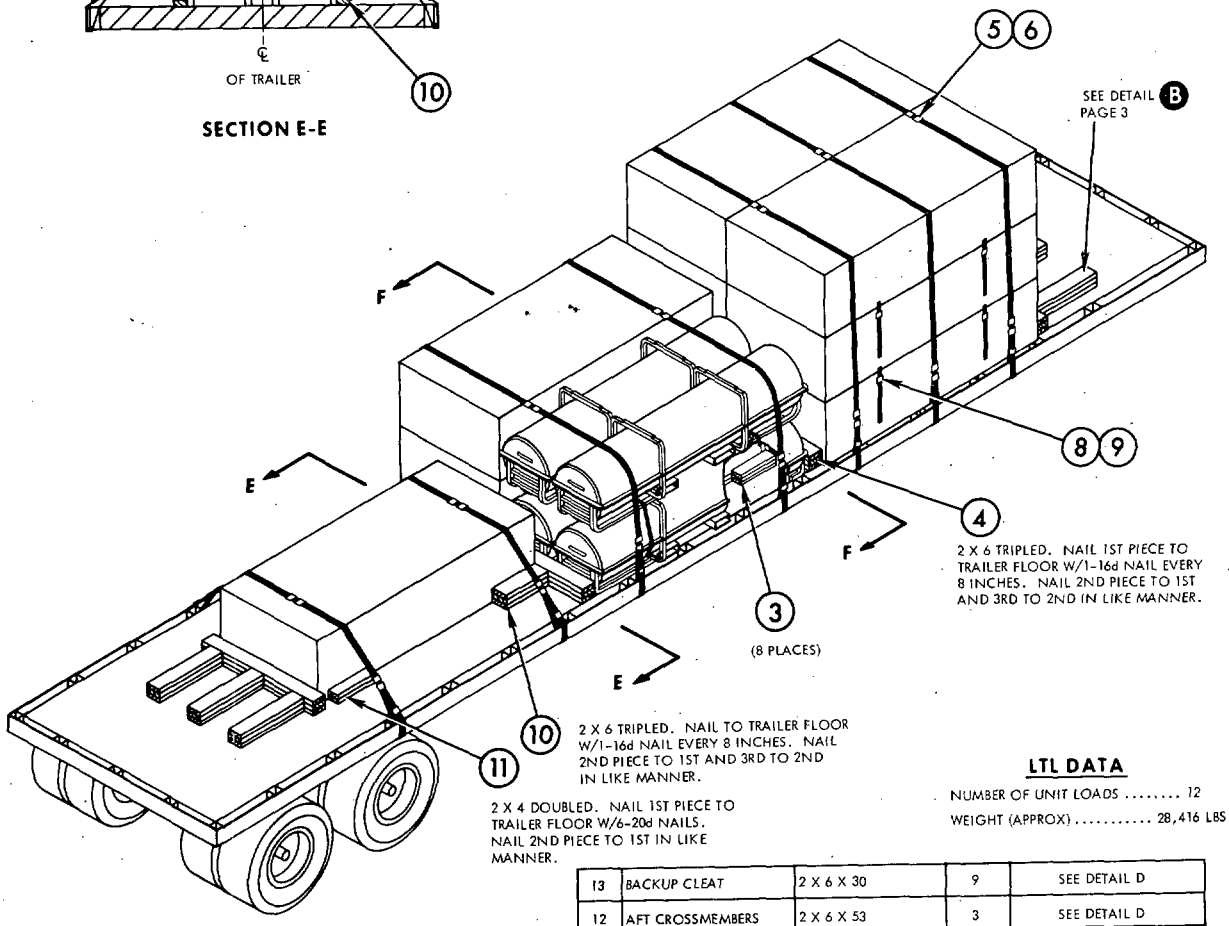
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SECTION E-E

**NOTE**

UNIT LOAD STRAPS
OMITTED FOR CLARITY




2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.

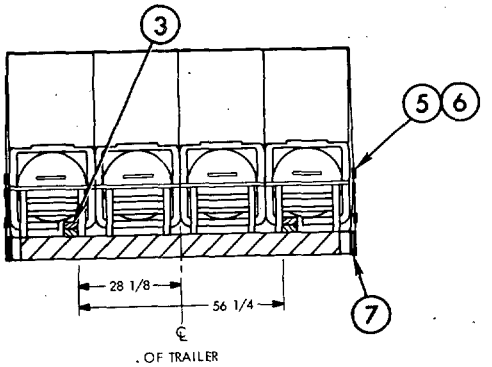
2 X 6 TRIPLED. NAIL TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.

2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-20d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

**LTL DATA**

NUMBER OF UNIT LOADS ..... 12
WEIGHT (APPROX) ..... 28,416 LBS

2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-20d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



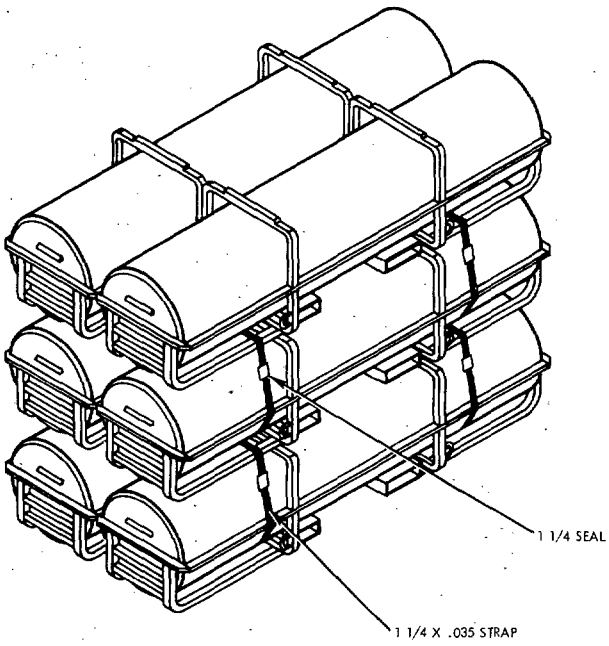

SECTION F-F

| PIECE NO. | DESCRIPTION | SIZE | NO. OF PIECES REQ'D | NAIL TO | NUMBER | SIZE |
|---|---|---|---|---|---|---|
| | LIST OF MATERIALS & NAILING DATA | | | | | |
| 13 | BACKUP CLEAT | 2 X 6 X 30 | 9 | SEE DETAIL D | | |
| 12 | AFT CROSSMEMBERS | 2 X 6 X 53 | 3 | SEE DETAIL D | | |
| 11 | SLEEPER | 2 X 4 X 18 | 4 | SEE FIELD NOTE | | |
| 10 | BACK UP | 2 X 6 X 20 | 6 | SEE FIELD NOTE | | |
| 9 | SEAL | FOR 1 1/4 IN. STRAP | 12 | - | - | - |
| 8 | UNITIZING STRAP | 1 1/4 X .035 X 13 FT 0 | 12 | - | - | - |
| 7 | PAD (STRAPPING) | 2 X .050 X 24 | 14 | - | - | - |
| 6 | SEAL | FOR 2 IN. STRAP | 56 | - | - | - |
| 5 | TIE DOWN STRAP | 2 X .050 X LENGTH TO SUIT | 7 | SEE LOAD STRAPPING, PAGE 2 | | |
| 4 | SEPARATOR | 2 X 6 X 92 | 6 | SEE FIELD NOTE | | |
| 3 | SLEEPER | 2 X 4 X 18 | 16 | SEE SECTION F-F | | |
| 2 | CLEAT | 2 X 6 X 30 | 15 | SEE DETAIL B, PAGE 3 | | |
| 1 | FWD CROSSMEMBER. | 2 X 6 X 92 | 3 | SEE DETAIL B, PAGE 3 | | |



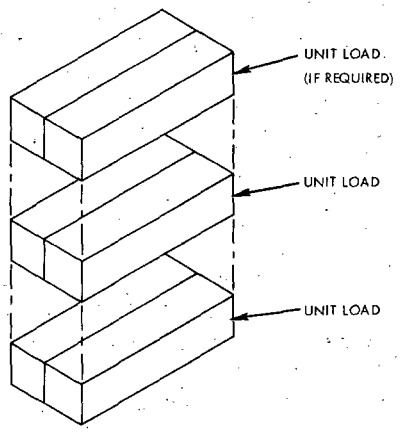
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# STACKING AND STRAPPING PROCEDURE




DETAIL G

NOTE
UNIT LOAD STRAPS
OMITTED FOR CLARITY




DETAIL H

1. USING A FORK LIFT TRUCK, STACK TWO (2) OR THREE (3) UNIT LOADS, AS REQUIRED, TOGETHER AS SHOWN IN DETAILS G & H. BE SURE TO ALIGN THE STACKING FEATURES.
2. THE MIDDLE CONTAINERS ARE STRAPPED TO THE BOTTOM CONTAINERS AND THE TOP CONTAINERS TO THE MIDDLE CONTAINERS USING 1 1/4 X .035 STEEL STRAPPING, TENSIONED AND SEALED WITH DOUBLE CRIMPED SEAL. THE STRAPS PASS OVER THE SKIDS OF THE UPPER CONTAINER AND UNDER THE BODY OF THE LOWER CONTAINER.

**CAUTION**

TENSION STRAPS SUFFICIENT TO INSURE INTEGRITY OF STACK. DO NOT OVER TENSION.

**WARNING**

THE STRAPPING TOGETHER OF UNIT LOADS TO FORM A UNIT OF CONTAINERS TWO (2) OR THREE (3) HIGH IS AUTHORIZED FOR LOADING AND UNLOADING OF TRAILERS ONLY. ANY MOVEMENT OF THESE LOADS IS LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER.



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